

Week 07: Inclusive Design & Accessibility

COMS 4170 • Spring 2026

Prof. Celeste Layne
Tuesday March 03, 2026



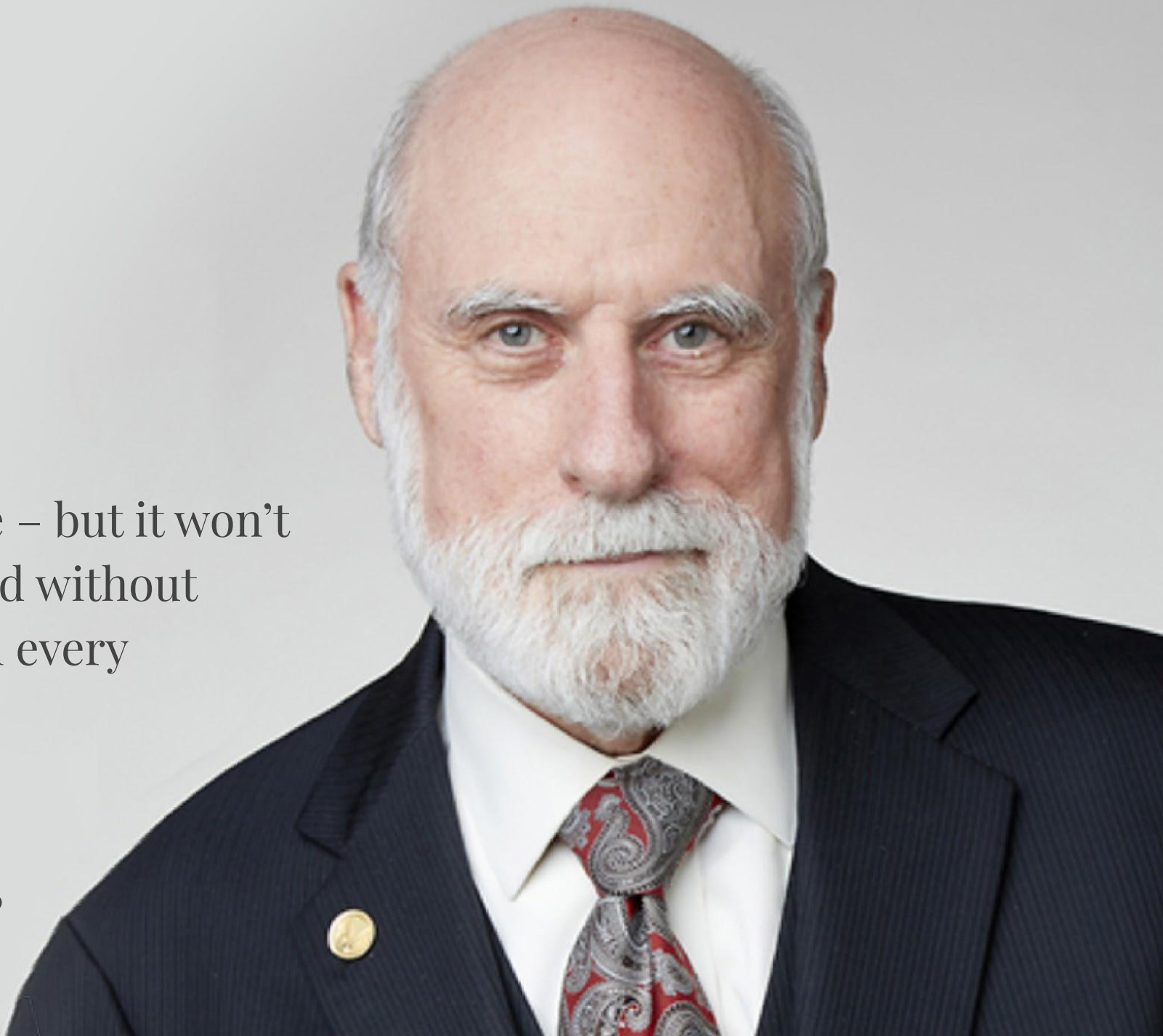
What is Accessibility?

1. The **qualities (or attributes)** that make an experience open to all. 2. A professional discipline aimed at achieving No. 1.



“The Internet is for everyone – but it won’t be until ... [it] can be accessed without limitation, at any time and in every language.”

– Vinton Cerf, co-founder of TCP/IP



Accessibility = Open to All

and there are laws in the USA that ensure the web is accessible



UC Berkeley student
Bruce Sexton sued
Target because its
website was
**inaccessible to the
blind.**

ars TECHNICA

POLICY —

Lawsuit over web site accessibility for the blind becomes class action

A federal judge has just certified a class action lawsuit against Target after ...

Blind shoppers win OK to sue Target

LAT ARCHIVES
OCT. 3, 2007 12 AM PT
FROM BLOOMBERG NEWS

Target Corp., the second-largest U.S. discount store chain, lost a bid Tuesday to dismiss a lawsuit claiming the company's website wasn't accessible to the blind.

U.S. District Judge Marilyn Hall Patel in San Francisco rejected Target's request to dismiss the case. She also certified the case as a class action, ruling that all legally blind people in the U.S. who have been denied access to services at Target stores because of deficiencies in the company's website can join the suit.

Target has failed to use "technologically simple and not economically prohibitive" code embedded in websites allowing the blind to use software that vocalizes the content, according to court filings by the National Federation of the Blind.

The group filed the suit on behalf of Bruce Sexton, a UC Berkeley student who claimed that he couldn't access some features of Target.com. "It was just gibberish for blind users trying to use the website," said Larry Paradis, a lawyer for the group.

starting at \$4.

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Top stories:

Web Feature



Transfer student Bruce Sexton has Target-ed independence for blind people as a major goal

By [Bonnie Azab Powell](#), NewsCenter | 20 March 2006

BERKELEY — Bruce "B.J." Sexton seems like a quiet, mild-mannered guy. But he can speak up when he has to. "People think I'm radical because I'm suing Target," says the third-year UC Berkeley student, who is president of the California Association of Blind Students and legally blind. "They don't try to understand what this is about."

He explains that he agreed to be a plaintiff, along with the National Federation of the Blind (NFB) and its California chapter, in a class-action suit filed February 7 against the discount retail giant only as a last resort. Starting in May 2005, the NFB and a Berkeley nonprofit law firm, Disability Rights Advocates, had been in active discussions with Target Corporation about making its website work with the screen-reading technologies used by the visually impaired, as many other business websites do. (To learn what's at issue with Target.com, see sidebar below.) When the talks broke down, the firm filed suit on behalf of the NFB and that of all blind people in California.

Bruce Sexton, an interdisciplinary studies major who is suing Target to make its website accessible for blind users. (Bonnie Azab Powell photos)

Men's spring fave

Prep for sunny days ahead

[ORDER ONLINE](#)[MENU](#)[COUPONS](#)[TRACKER](#)[REWARDS](#)[VIEW YOUR LOCAL DOMINO'S](#)[SIGN IN & EARN REWARDS](#)

Guillermo Robles, who is blind, claimed in U.S. District Court in California that the pizza maker violated the federal disability requirements because he couldn't order a pizza on his iPhone:

The website didn't work with his screen-reader software.

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MONEY

Victory for disability advocates: Supreme Court won't hear Domino's Pizza accessibility case

Kelly Tyko USA TODAY

Published 9:56 p.m. ET Oct. 7, 2019 | Updated 3:04 p.m. ET Oct. 8, 2019

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Domino's

Domino's store Domino's Pizza

The U.S. Supreme Court's decision to not hear Domino's petition on whether its website is accessible to the disabled is considered a loss for the pizza giant and a win for disability advocates.

Accessibility Today

Internet inventor: Make tech accessibility better already

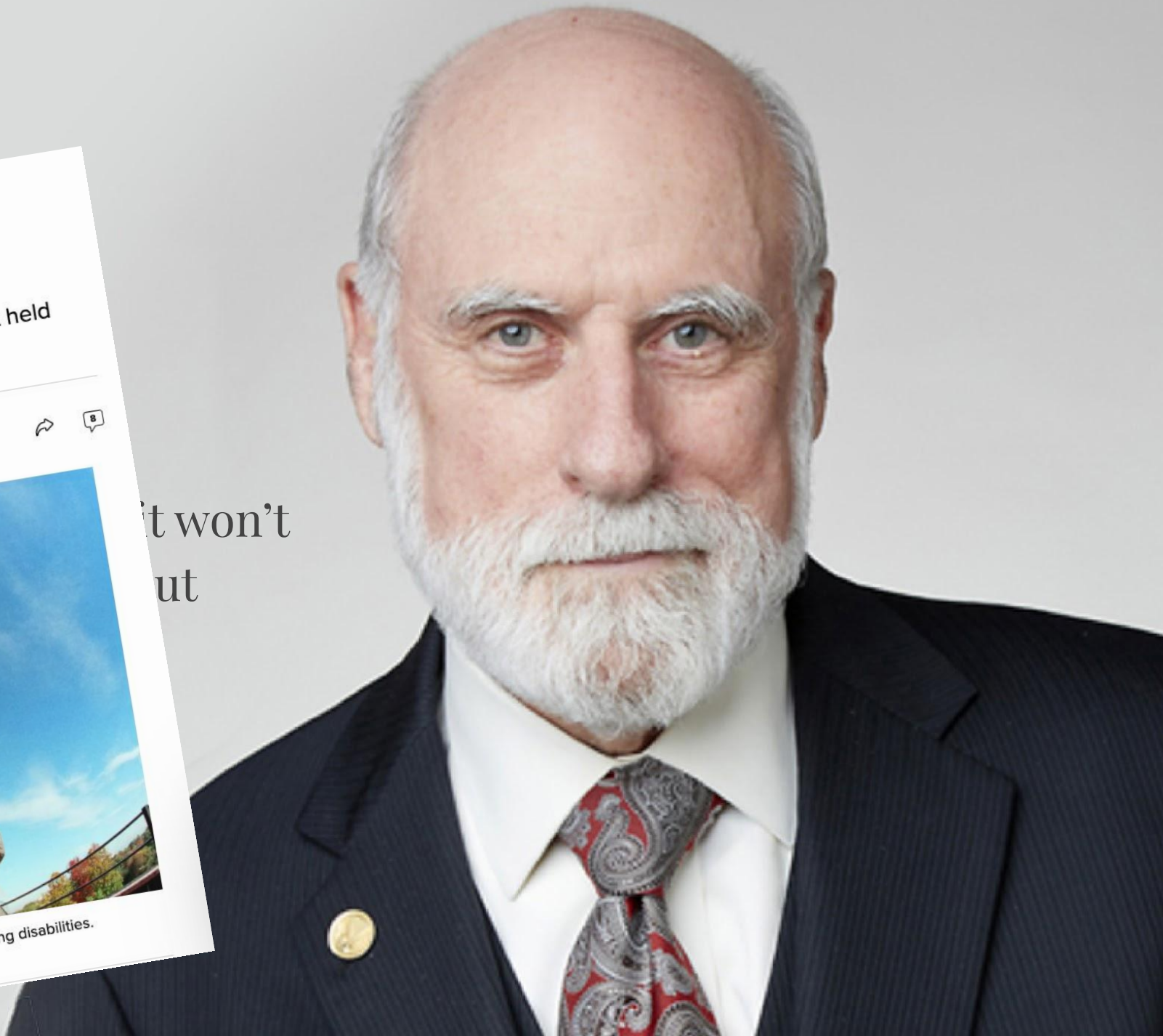
Vint Cerf says "it's almost criminal" that programmers aren't held accountable to design with disabilities in mind.

Joan E. Solsman April 10, 2017 5:00 a.m. PT



Both Vint Cerf, known as a "father of the Internet," and his wife have hearing disabilities.

it won't
ut



World Health Organization redefined disability

Disability is not a property of a person,
but a **property of a system**.

It reflects the interaction between features of a person's body and features of the society in which he or she lives.

Disability is part of being human in the world.



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What assumptions do we make about people?



What assumptions do we make about users?



- Right-handed
- Fine motor control to move mouse around
- Fine motor control for scroller and ball motion
- Enough strength to press the buttons/move the ball
- Sensitivity to find smaller buttons

What assumptions do we make about users?



- They can see well
 - Distinguish between colors
 - Distinguish between small icons
 - Distinguish between small letters (when typing)
- They can hold it in their hand
 - Arm mobility
 - Grip strength
 - Fine motor control
- They have high control over their fingers

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Visual

Blindness



Ways that websites fail the visually-impaired

- Poorly structured HTML
- Images with no meaningful alt text
- Poorly structured forms
- Repetitive items that cannot be skipped
- Features that require vision, or where the fallback is poorly implemented



How does a blind person use a computer?



How does a blind person use a computer?

- Screen reader says out loud what is on the screen
- Gestures are generally not location specific
- Feedback on location specific gestures corrects for error



Ways that websites fail the visually-impaired

Heading Structure

Headers should descend logically and should be used for section headers.

```
<!-- Example 2 -->
<!-- Be as explicit as possible when using tags and always include headings for main titles -->
<!-- Do This -->

<h1>Special Pizza NYC</h1>
  <h2>About Special Pizza</h2>
    <h3>Our Story</h3>
    <h3>Our Founder</h3>
  <h2>Why Our Pizza Is Special</h2>
    <h3>Our Ingredients</h3>
    <h3>Our Chef</h3>

<!-- Not This -->
<div>Special Pizza NYC</div>
<div>About Special Pizza</div>
<div>Why Our Pizza is Special</div>
```



Ways that websites fail the visually-impaired

Using Accessible Rich Internet Applications (ARIA)

ARIA roles provide screen reader users with more information, context and greater interactivity. ARIA roles are relevant to making sure readers can perceive content.

You can add an ARIA role by simply assigning a role attribute to a tag:

```
<div role="banner">
```



Ways that websites fail the visually-impaired

Alt tags

Alt tags are added to images in order to help users understand what they are unable to **perceive** or **interact** with, particularly if they have low or no vision.

```
<!-- Example 3 -->  
<!-- Add alternate text descriptions to images -->  
<!-- Do This -->  
  
  
  
<!-- Not This -->  
  

```



Ways that websites fail the visually-impaired

Forms

```
<form>
```

```
  <label for="first_name">Enter your first name</label>
```

```
  <input type="text" id="first_name">
```

```
</form>
```



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Visual

Low Vision and Color Blindness



Ways that websites fail the visually-impaired

Accessible Text

- Relative units
- Stop using pixels
- Raise your base font size
- Line height
- Letter spacing

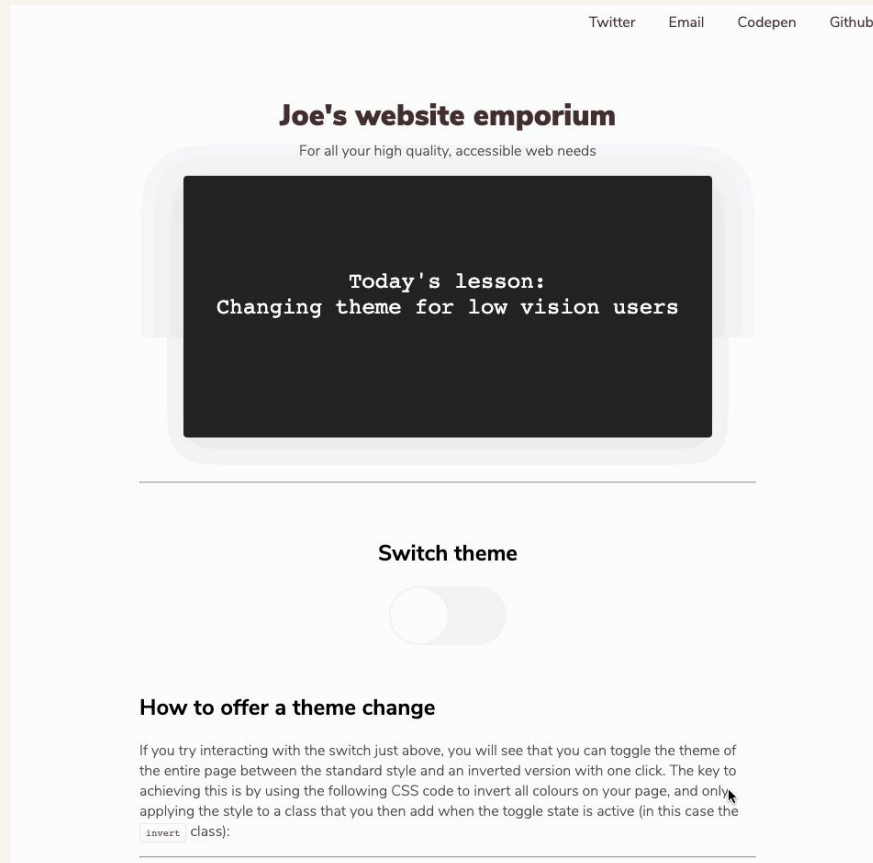


Ways that websites fail the visually-impaired

Contrast Ratio

Depending on the browser supported, CSS has a filter property available that can create a whole range of different visual effects.

```
body.invert{  
  
    filter: invert(100%);  
  
}
```



Ways that websites fail the visually-impaired

Color Blindness

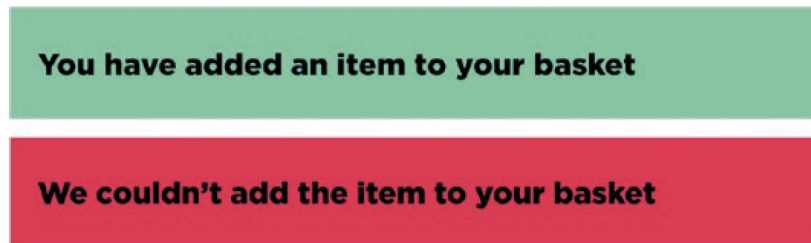


Figure 3-10. Two alerts, one positive and one negative, both where colour is the primary way in which status is displayed

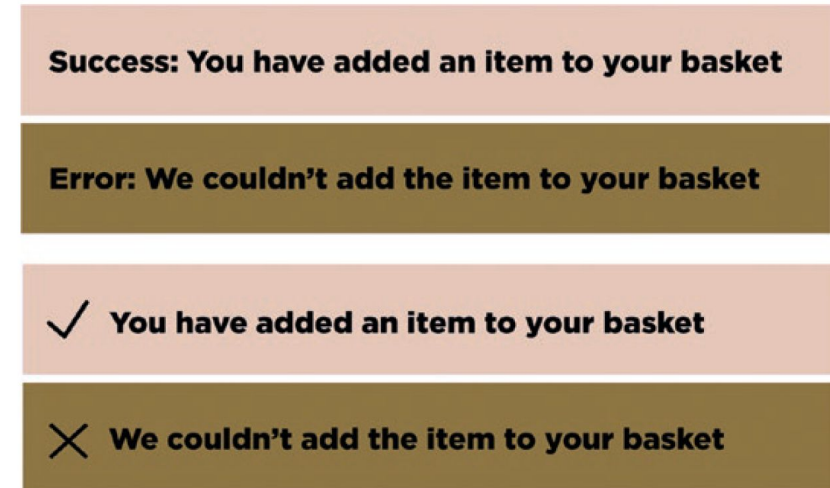


Figure 3-12. The status messages now use either "Success" and "Error", or a tick and a cross icon, so the meaning of the message doesn't rely on colour

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Audio



Ways that websites fail the hearing-impaired

- Videos with no captioning
- Videos with poor captioning
- Interactive features with no visual alerts
- Poor quality live feeds



Ways that websites fail the hearing impaired

<track> element

HTML5 allows us to add subtitles for a video using the <track> tag:

```
<video controls>
  <source src="videos/video.mp4" type="video/mp4">
  <source src="videos/video.webm" type="video/webm">
  <track label="English"
        kind="subtitles"
        srclang="en"
        src="captions/vtt/video-en.vtt"
        default>
</video>
```



Ways that websites fail the hearing impaired

<audio> tag and captions

```
/* CSS */  
.audio {  
  height: 120px;  
}  
  
WEBVTT  
  
00:00:08.100 --> 00:00:10.500 line:1  
I'm not the monster here. You are.
```



Figure 5-9. A custom audio player using the <video> tag, a fixed height, and positioned captions

```
<video class="audio" controls>  
  <source src="media/audio.mp4" type="audio/mpeg">  
  <track label="English" kind="subtitles" srclang="en" src="vtt/audio.vtt" default/>  
</video>
```

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Physical



Ways that websites fail the physically impaired

Keyboard-only navigation



Figure 4-1. A picture of a button and then that same button with keyboard focus style applied

Have you ever tried using a website without a mouse?

The most common method of keyboard-only navigation relies on the Tab key, which moves the user sequentially between elements on a page. When you're focused on an element using the keyboard, by default an outline appears to reflect that.

Ways that websites fail the physically impaired

Focus styles



Imagine you're a keyboard-only user and you're aiming to interact with one of these menu items. In the first example, you're left to guess which link you're on and could potentially end up heading to pages you weren't trying to navigate to.

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Cognitive



Ways that websites fail the **cognitively impaired**

Clear iconography



Figure 6-4. A “speech bubble” icon

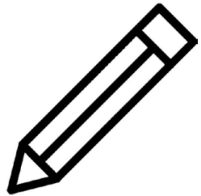


Figure 6-5. A “pencil” icon

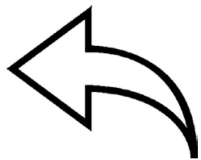


Figure 6-6. An icon of an arrow, pointing upward and left

Speech bubble. it means **view post and comments** in an instagram feed but means **add a comment** when a user is on an individual instagram post.

Pencil icon. Used inside a square to mean **create a note** in Apple Notes.

Arrow icon. Used to represent **share a post** on LinkedIn.

Ways that websites fail the cognitively impaired

Autocomplete

Search for a country:



A screenshot of a search input field with the text 'cro' entered. Below the input field is a dropdown menu with two options: 'Croatia' and 'Micronesia (Federated States of)'. The 'Croatia' option is highlighted in blue.

cro
Croatia
Micronesia (Federated States of)

Figure 6-15. Strict search accurately returning “Croatia” when the user has entered “cro”

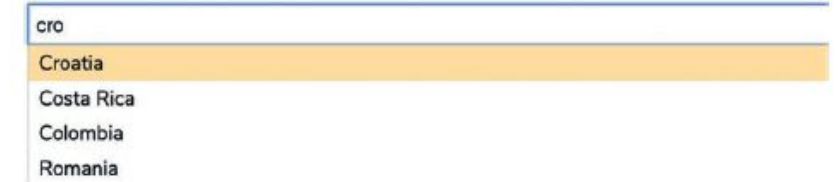
Strict Search

Relies on an exact match e.g. jQueryUI Autocomplete.

Fuzzy Matching

Doesn't rely on an exact match

Search for a country or ISO code:

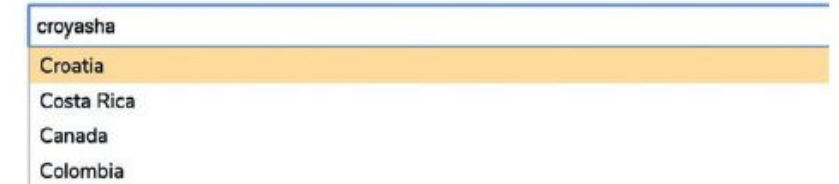


A screenshot of a search input field with the text 'cro' entered. Below the input field is a dropdown menu with four options: 'Croatia', 'Costa Rica', 'Colombia', and 'Romania'. The 'Croatia' option is highlighted in orange.

cro
Croatia
Costa Rica
Colombia
Romania

Figure 6-17. Fuzzy search accurately returning “Croatia” when the user has entered the start of the word – “cro” – the same as strict search

Search for a country or ISO code:

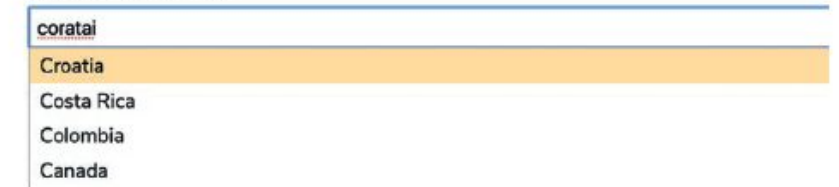


A screenshot of a search input field with the text 'croyasha' entered. Below the input field is a dropdown menu with five options: 'Croatia', 'Costa Rica', 'Canada', and 'Colombia'. The 'Croatia' option is highlighted in orange.

croyasha
Croatia
Costa Rica
Canada
Colombia

Figure 6-18. Fuzzy search accurately returning “Croatia” when the user has entered an incorrect spelling based on how the word is said – “croyasha”

Search for a country or ISO code:



A screenshot of a search input field with the text 'coratai' entered. Below the input field is a dropdown menu with five options: 'Croatia', 'Costa Rica', 'Colombia', and 'Canada'. The 'Croatia' option is highlighted in orange.

coratai
Croatia
Costa Rica
Colombia
Canada

Figure 6-19. Fuzzy search accurately returning “Croatia” when the user has entered all of the correct letters for the word but in the wrong order – “Coratai”

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Inclusive design focuses on individuals



The myth of average



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News · Insight

When U.S. air force discovered the flaw of averages

By **TODD ROSE**
Sat., Jan. 16, 2016



In the late 1940s, the United States air force had a serious problem: its pilots could not keep control of their planes. Although this was the dawn of jet-powered aviation and the planes were faster and more complicated to fly, the problems were so frequent and involved so many different aircraft that the air force had an alarming, life-or-death mystery on its hands. “It was a difficult time to be flying,” one retired airman told me. “You never knew if you were going to end up in the dirt.” At its worst point, 17 pilots crashed in a single day.



Designing for the Average

Pros

- Only have to design one system
 - Faster and simpler
- Maybe you can model after yourself
 - Need fewer test users
- Almost always cheaper

Cons

- Limit who can use your system
- Might be imperfect for everyone
- Might make incorrect assumptions about what average means
- Might never learn how to design for others

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Situational impairments



World Health Organization redefined disability

Disability is not a property of a person,
but a **property of a system**.

It reflects the interaction between features of a person's body and features of the society in which he or she lives.

Disability is part of being human in the world.



What are some situational impairments of this system?



- Only using one hand
- Distracted (riding a bike)
- Wearing a long coat
- Diminished hearing

What are some situational impairments of this system?



- Only using one hand
- Holding a baby rattler
- Distracted

What are some situational impairments of this system?



- Diminished hearing
- Distracted
- Low lighting
- Cognitive impairment

What is Inclusive Design?

A **design methodology** that enables and draws on the full range of human diversity. Most importantly, this means including and learning from people with a range of perspectives.

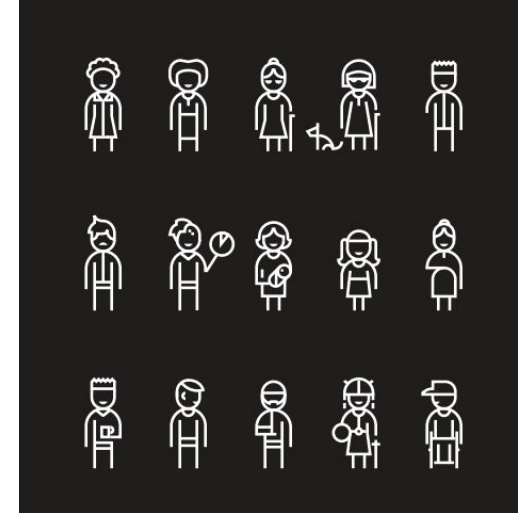


Inclusive Design



Who we design for

If we use our own abilities and biases as a starting point, we end up with products designed for people of a specific gender, age, language ability, tech literacy, and physical ability. Those with specific access to money, time, and a social network.



Who gets excluded

When it comes to people, **there's no such thing as "normal."** The interactions we design with technology depend heavily on what we can see, hear, say, and touch. Assuming all those senses and abilities are fully enabled creates the potential to ignore much of the range of humanity.

Inclusive Design

3 Principles of Inclusive Design

1

Recognize
exclusion

2

Learn from
diversity

3

Solve for
one, extend
to many



Recognize Exclusion



One arm



Arm injury



New parent

Situational impairments

Sometimes exclusion is **situational**. e.g. inability to hear in a loud crowd; trying to see after looking into a bright light.

Activity limitations

Context affects how people interact with the world: e.g. a new parent spending their day doing one-handed tasks.

Restrictions on participation

Sometimes exclusion is **temporary**. e.g. ordering dinner in a foreign country; wearing a cast on your arm.

Learn from Diversity

The insight is in the adaptation

When experiences don't serve people the way they should, people adapt. Insights come when we understand those adaptations.

Mobile technology puts in focus questions like: Are we forced to adapt to technology, or is technology adapting to us?

Learning how people adapt to the world around them means spending time understanding their experience from their perspective.

Solve for one, extend to many



Permanent **26K**



Temporary **13M**



Situational **8M**

The beauty of constraints

High-contrast screen settings were initially made to benefit people with vision impairments. Today, many people benefit from high-contrast settings when they use a device in bright sunlight.

Different people benefit

By designing for someone with a permanent disability, someone with a situational limitation can also benefit.

More people benefit

In the United States, 26,000 people a year suffer from loss of upper extremities. But when we include people with temporary and situational impairments, the number is greater than 20M.

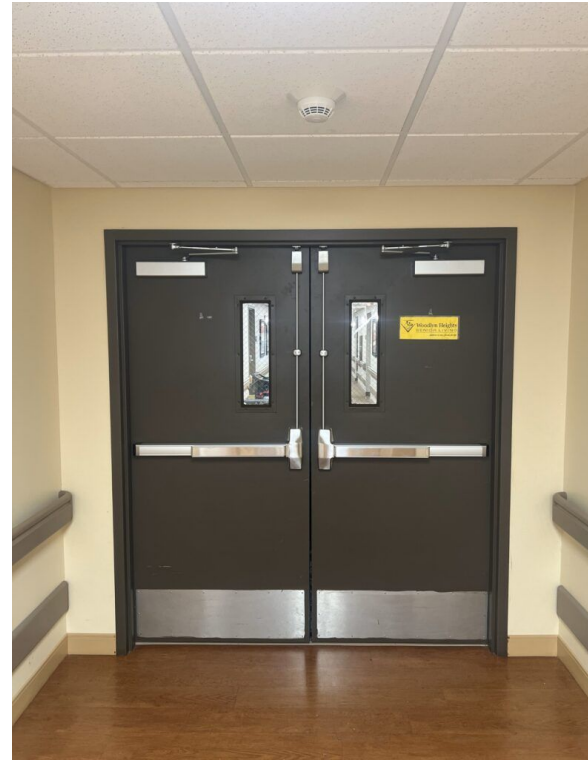
Designed for wheelchairs. Who else benefits?



- Diminished hearing
- Distracted
- Low lighting
- Cognitive impairment



Designed for fire safety. Who else benefits?



- People in a rush
- People holding two cups of coffee
- People who cannot read
 - children
 - elderly
 - non-english speakers

Designed for arthritis. Who else benefits?



- Anyone who has ever needed to peel anything.



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Summary



Accessibility Today

World Health Organization

Disability is part of being human.

It is not a personal attribute but context dependent...reflecting the interaction between features of a person's body and features of the society in which he or she lives.



The myth of average



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Midterm Project



Midterm Project

Due **Friday 03/13 at 11:59PM**

10 days from now

*It's worth **10% of your grade.***

Midterm:

Usable CRUD Application

Warm Up Due: Friday Mar 06, 2026 11:59 PM on Courseworks
Main Due: Monday Mar 09, 2026 11:59 PM on Courseworks
BOTH will be accepted as on time until 8:00 AM the next morning

Overview:

Welcome to your midterm – it displays cumulative knowledge of the things learned in all the lectures in the first half of the class. In this class it is worth 10% of your final grade. Although it has the same structure as a homework assignment, treat it with the same urgency as an in-class midterm. Just like a homework assignment, you are still allowed to Google things, use ChatGPT, come to office hours, work with others, and seek other forms of help.

You **cannot use a late pass** on this midterm assignment. If you have an emergency situation that requires an extension, please contact your advising dean and have them reach out to me.

Additionally, the staff will not hold office hours or answer piazza questions during the week of Spring Break.

Think of this Midterm as the first of two portfolio pieces that demonstrate your technical and design skills, and your ability to put them together for a polished final product. It's worth putting in some time to refine the design, and make a decent video. We highly recommend getting feedback on your design from a TA during office hours. Even a little feedback makes products much better.

Warm Up: Visual Information Design

Visual Information Design Lebowski

Search in Presentation

Share

Office Update To keep up-to-date with security updates, fixes, and improvements, choose Check for Updates. Check for Updates

Visual Information Design

1

2

3

4

5

6

The Big Lebowski 1998



Genres: buddy, screwball, indie, stoner

Directors: [Joel Coen](#), [Ethan Coen](#)

Cast: [Jeff Bridges](#), [John Goodman](#), [Julianne Moore](#)

★★★★★

When 'The Dude' Lebowski is mistaken for a millionaire Lebowski, two thugs urinate on his rug to coerce him into paying a debt he knows nothing about. While attempting to gain recompense for the ruined rug from his wealthy counterpart, he accepts a one-time job with high pay-off. He enlists the help of his bowling buddy, Walter, a gun-toting Jewish-convert with anger issues. Deception leads to more trouble, and it soon seems that everyone from porn empire tycoons to nihilists want something from The Dude.",

Budget: \$15,000,000

Slide 6 of 6 English (United States)

Notes Comments

110%



Purpose and User Background

- What data will your website allow users to search?
 - **Example:** “Crime Comedy Films.” (You cannot not use this answer).
- What is an example of a person who would want to search this and why?
 - **Example:** A college student who saw The Big Lebowski and is now fascinated by the genre of crime comedies and wants to find more to watch.
- What information on this page should help the user issue a new search? And why?
 - **Example:** The genres, directors and actors are searchable because they can help the user find similar movies they may like.



Three (3) Conceptual Groups

```
data = {  
    "1": {  
        id: 1,  
        title: "The Big Lebowski",  
        poster: "https://m.media-amazon.com/images/lebowski.jpg",  
        year: 1998,  
        summary: "When 'The Dude' Lebowski is mistaken for a millionaire Lebowski, two thugs urinate on his rug to coerce him into paying a debt he knows nothing about. While attempting to gain recompense for the ruined rug from his wealthy counterpart, he accepts a one-time job with high pay-off. He enlists the help of his bowling buddy, Walter, a gun-toting Jewish-convert with anger issues. Deception leads to more trouble, and it soon seems that everyone from porn empire tycoons to nihilists want something from The Dude.",  
        director: ["Joel Coen", "Ethan Coen"],
```



The Big Lebowski

1998



Genres

buddy

screwball

indie

stoner

Directors

[Joel Coen](#), [Ethan Coen](#)

Cast:

[Jeff Bridges](#)

[John Goodman](#)

[Julianne Moore](#)



When 'The Dude' Lebowski is mistaken for a millionaire Lebowski, two thugs urinate on his rug to coerce him into paying a debt he knows nothing about. While attempting to gain recompense for the ruined rug from his wealthy counterpart, he accepts a one-time job with high pay-off. He enlists the help of his bowling buddy, Walter, a gun-toting Jewish-convert with anger issues. Deception leads to more trouble, and it soon seems that everyone from porn empire tycoons to nihilists want something from The Dude."

Budget: \$15,000,000



Concept:
What is it?

Most important:
Title is most important.

Tools:
Size, contrast, position

The Big Lebowski

1998



Genres

buddy

screwball

indie

stoner

Directors

[Joel Coen](#), [Ethan Coen](#)

Cast:

[Jeff Bridges](#)

[John Goodman](#)

[Julianne Moore](#)

Concept:
Do I want to watch it?

Most important:
Stars.

Tools:
Size, Image, color, position



When 'The Dude' Lebowski is mistaken for a millionaire Lebowski, two thugs urinate on his rug to coerce him into paying a debt he knows nothing about. While attempting to gain recompense for the ruined rug from his wealthy counterpart, he accepts a one-time job with high pay-off. He enlists the help of his bowling buddy, Walter, a gun-toting Jewish-convert with anger issues. Deception leads to more trouble, and it soon seems that everyone from porn empire tycoons to nihilists want something from The Dude.",

Budget: \$15,000,000

Concept: Help me find more like this.

Most important: Genres you can click on

Tools: Size, contrast, position



Main Assignment: Complete CRUD Application

Functionality

- Homepage
- View Item
- Search
- Add Item
- Edit Item

Visual Design

- Visual Information Hierarchy
- Colors
 - base, accent, light grey, dark grey
- Accessibility
 - alt text



Next week: No lecture

- Office hours – check the website!
- If you want to see HW6 Solutions, ask a TA! We want to walk you through it and help you understand, so you can build on it.



After Spring Break, we switch to group projects

- We will be learning and practicing human-centered design.
- On Tuesday, you'll meet in your TA group:
 - **In-person section:** Your TA will tell you a time and place
 - **Zoom section:** Your TA will reach out to you with a link.
 - **CVN:** Your TA will reach out to you. We may find a time that works for everyone, we may give feedback asynchronously



